16_Multiple_Imputation_Diagnostics

Anne Gaule

2026-09-14

Contents

1) Check logged events **3**

2) Checking Distributions of derived scores **4**

1) Child Summary scores for each timepoint . . . 5

ASB T1 . . . 5

ASB T2 . . . 6

ASB T3 . . . 7

PSB T1 . . . 8

PSB T2 . . . 9

PSB T3 . . . 10

2) Teacher sumary scores for each timepoint . . . 11

ASB T1 . . . 11

ASB T2 . . . 12

ASB T3 . . . 13

PSB T1 . . . 14

PSB T2 . . . 15

PSB T3 . . . 16

3) Kernel density plots comparing observed and imputed distributions for the individual items comprising each scale **17**

1) Teach ASB D1 . . . 17

2) Teach ASB D2 . . . 18

3) Teach ASB D3 . . . 19

4) Teach PSB D1 . . . 20

5) Teach PSB D2 . . . 21

6) Teach PSB D3 . . . 22

7) Child ASB D1 . . . 23

8) Child ASB D2 . . . 24

9) Child ASB D3 25
10) Child PSB D1 26
11) Child PSB D2 27
12) Child PSB D3 28

1) Check logged events

Only 12 logged events - only 12 predictor removals occurred across the full imputation run, indicating the resulting imputation models were numerically stable.

```
#only 7 warnings or errors
check <- imputed10$loggedEvents

#Check convergence
check %>% group_by(dep, out) %>% summarise(n=n())
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by dep and out.
## i Output is grouped by dep.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(dep, out))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

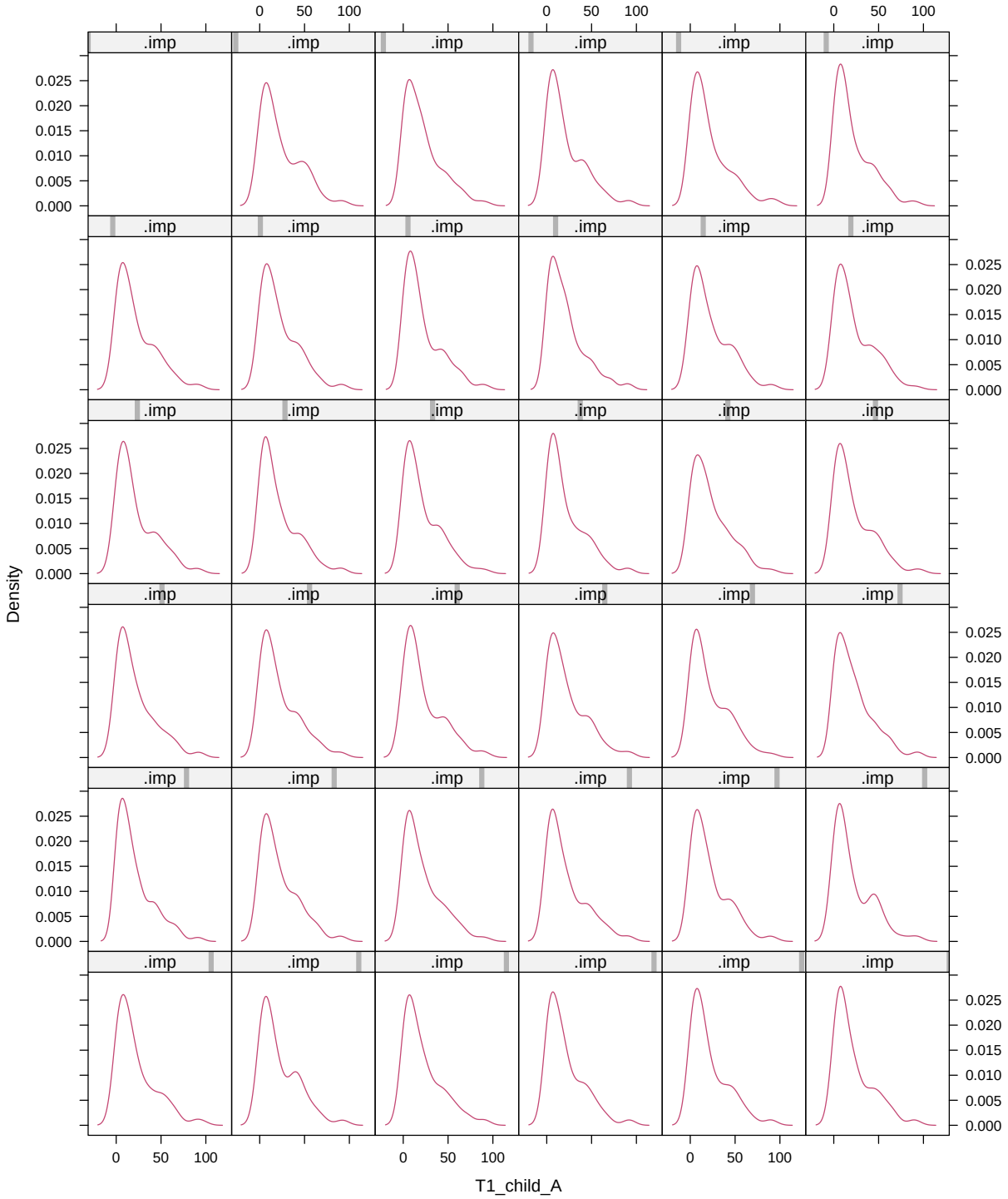
```
## # A tibble: 13 x 3
## # Groups:   dep [2]
##    dep       out           n
##    <chr>     <chr>     <int>
##  1 D2_5_A_Q2 D1_4_A_Q2     1
##  2 D3_4_A_Q2 D1_2_A_Q5     3
##  3 D3_4_A_Q2 D1_4_A_Q2     1
##  4 D3_4_A_Q2 D1_5_A_Q3     3
##  5 D3_4_A_Q2 D2_1_A_Q2     1
##  6 D3_4_A_Q2 D2_2_A_Q5     2
##  7 D3_4_A_Q2 D2_3_A_Q1     1
##  8 D3_4_A_Q2 D2_3_A_Q2     3
##  9 D3_4_A_Q2 D2_5_A_Q4     1
## 10 D3_4_A_Q2 D2_5_A_Q6     2
## 11 D3_4_A_Q2 D3_1_A_Q4     2
## 12 D3_4_A_Q2 D3_2_A_Q4     3
## 13 D3_4_A_Q2 D3_2_A_Q5     4
```

2) Checking Distributions of derived scores

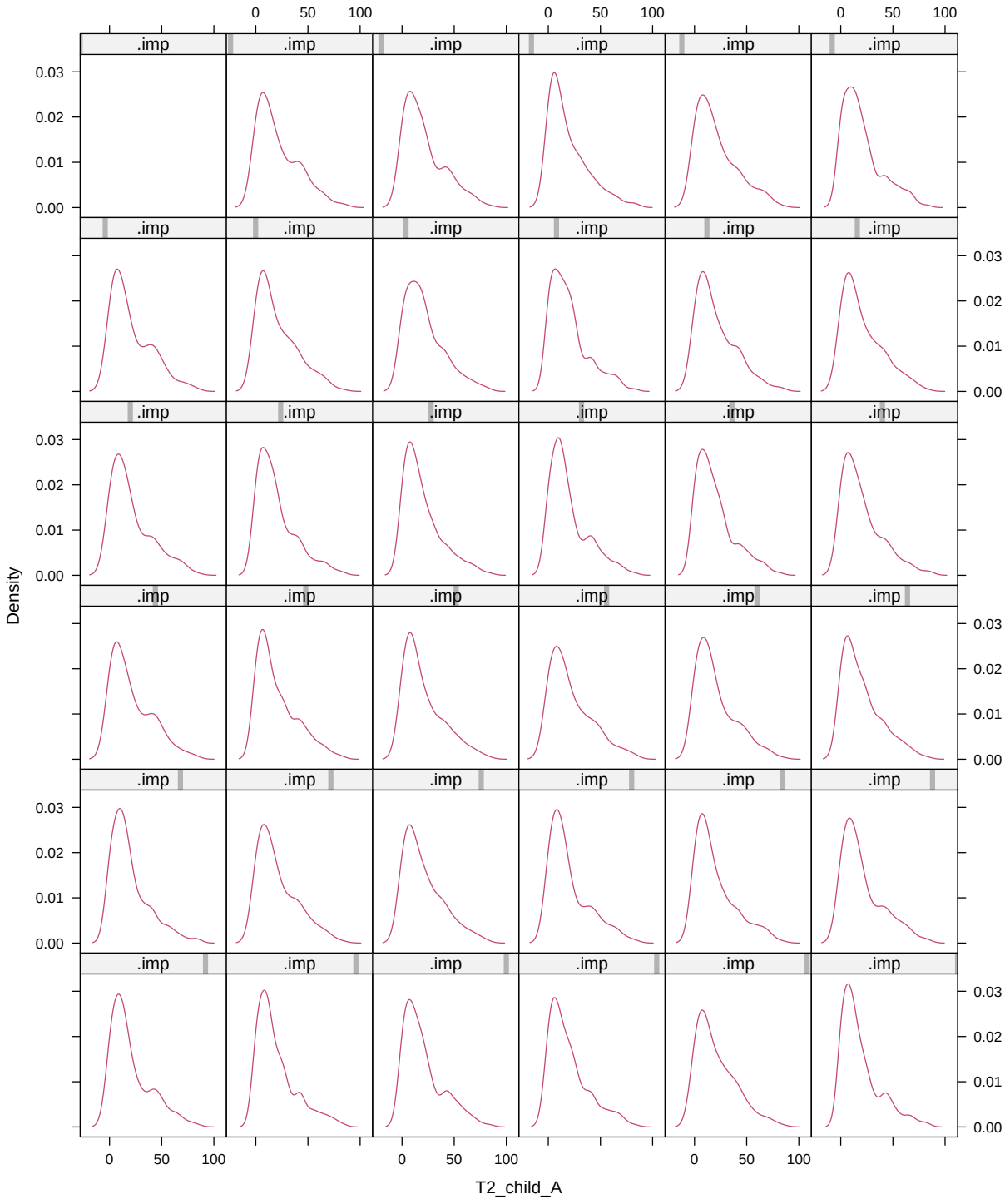
Because scale scores were derived from imputed item-level responses rather than imputed directly, distributions of the derived scores were compared across the 35 imputed datasets to confirm stability.

1) Child Summary scores for each timepoint

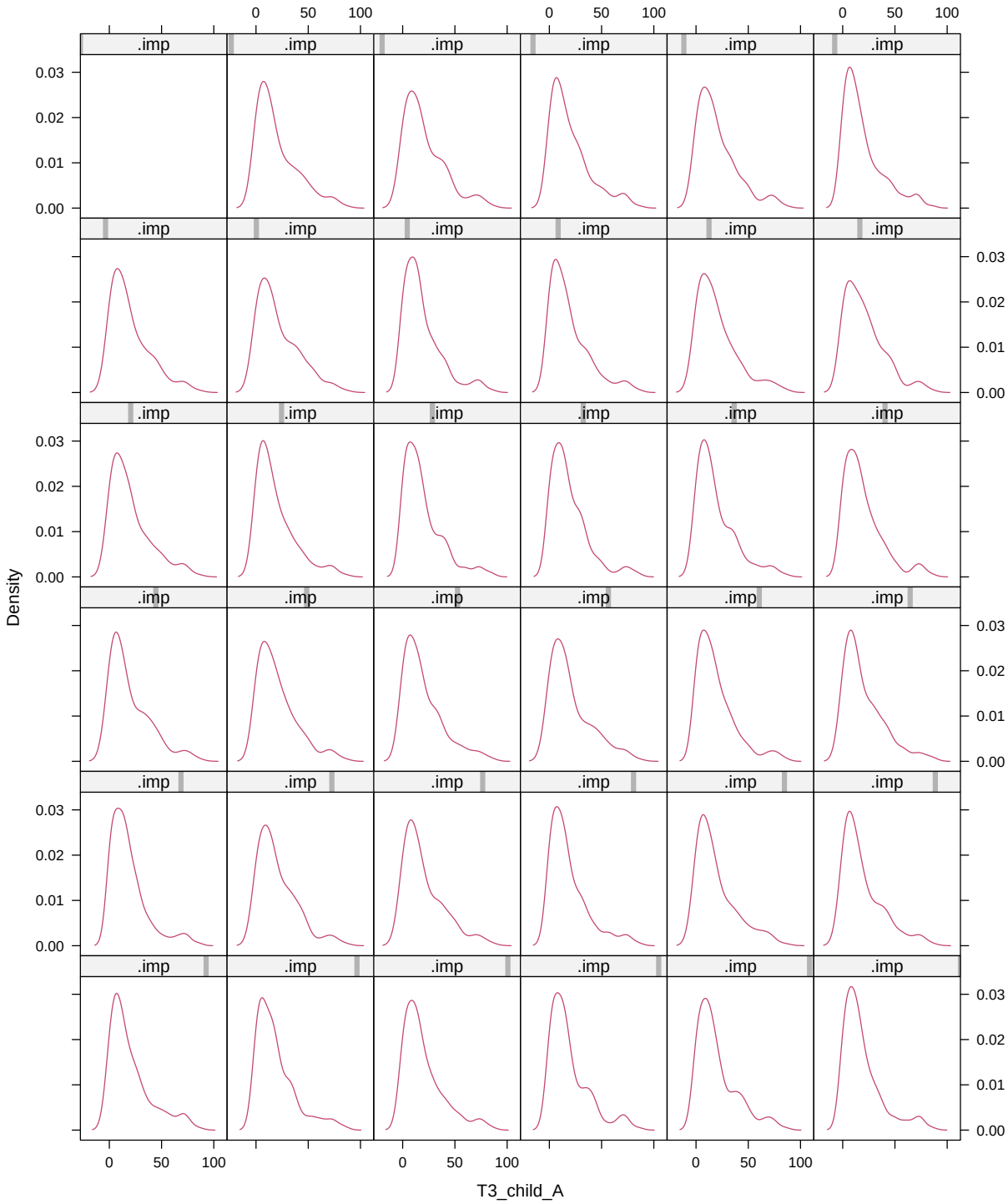
ASB T1



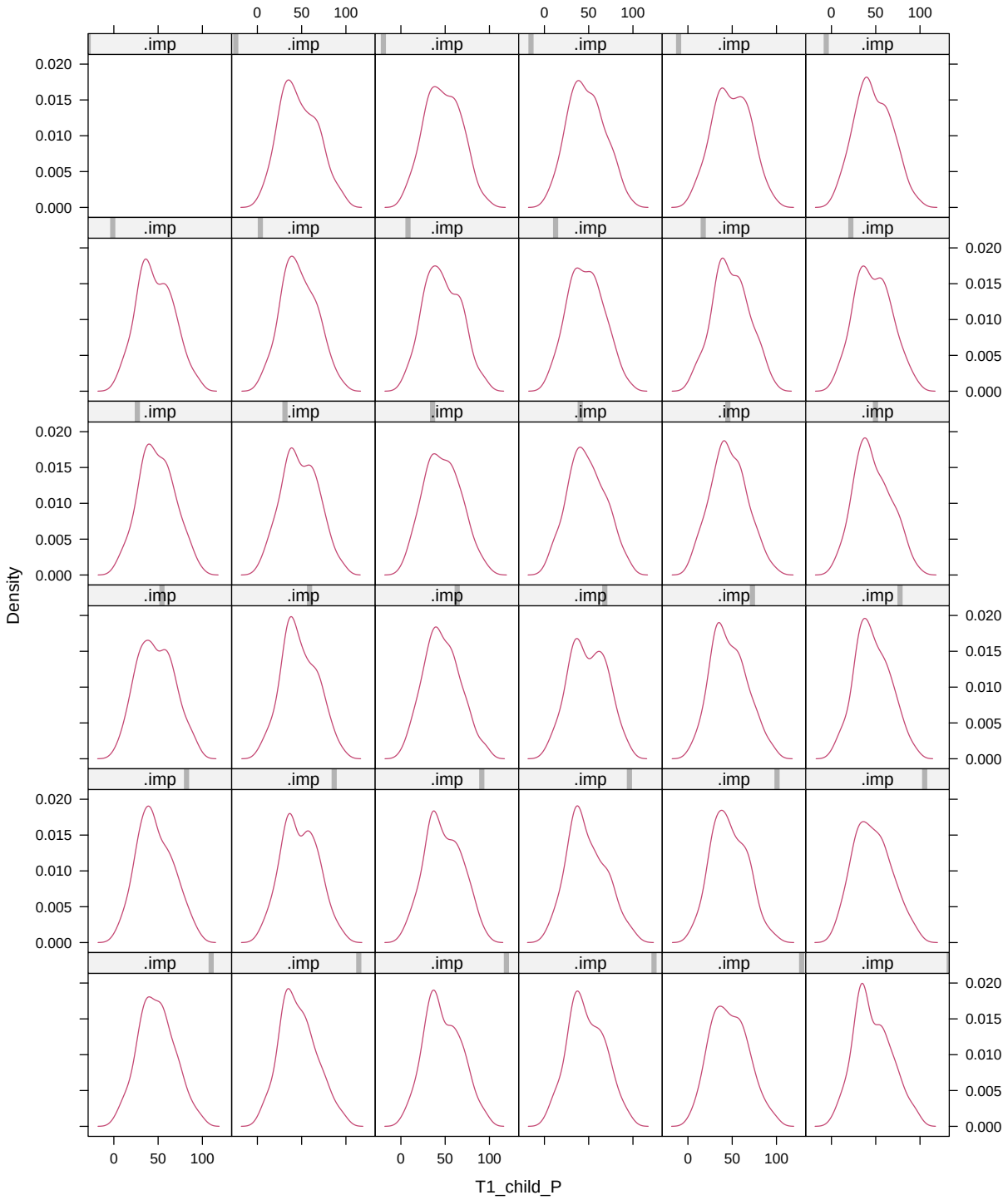
ASB T2



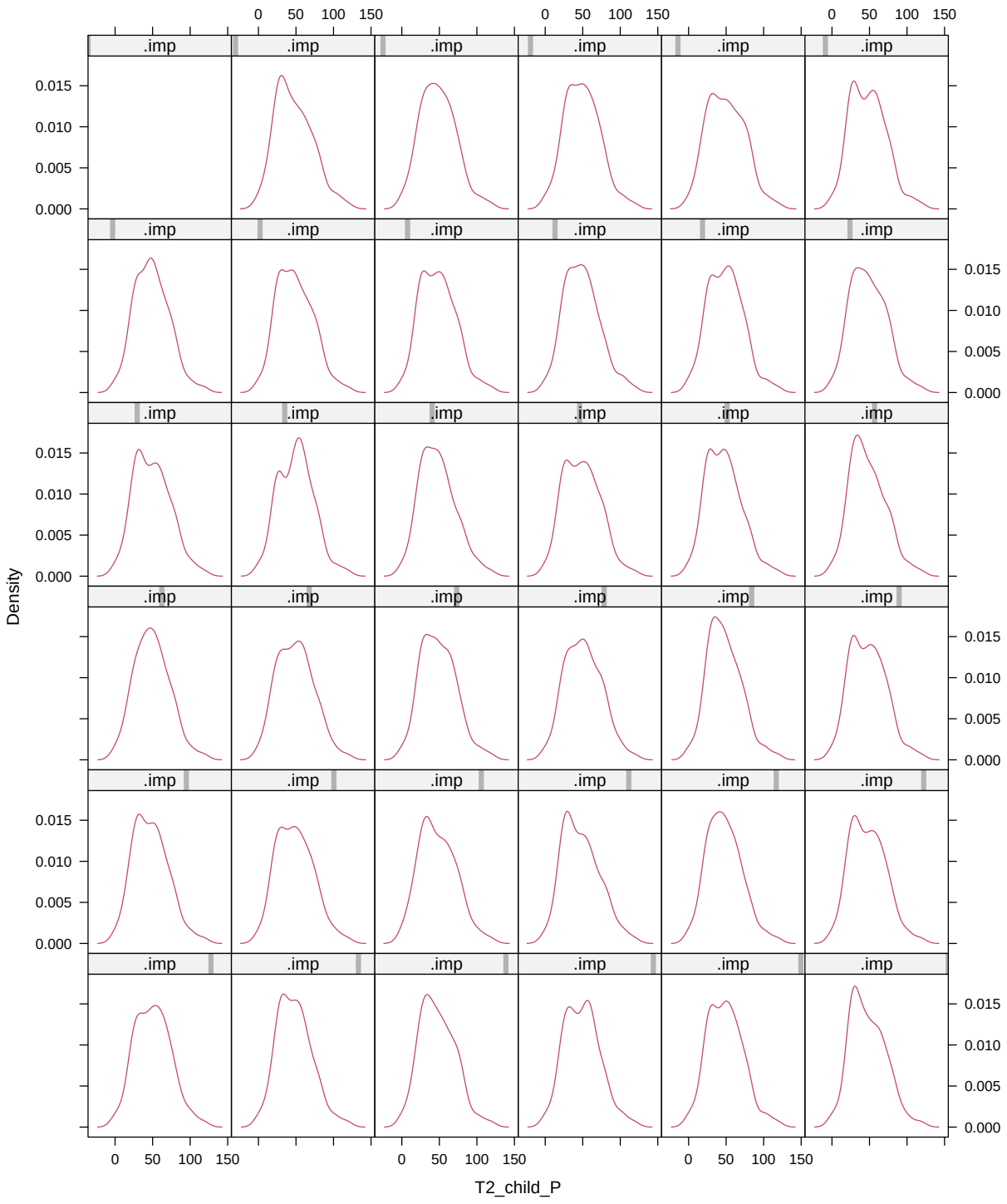
ASB T3



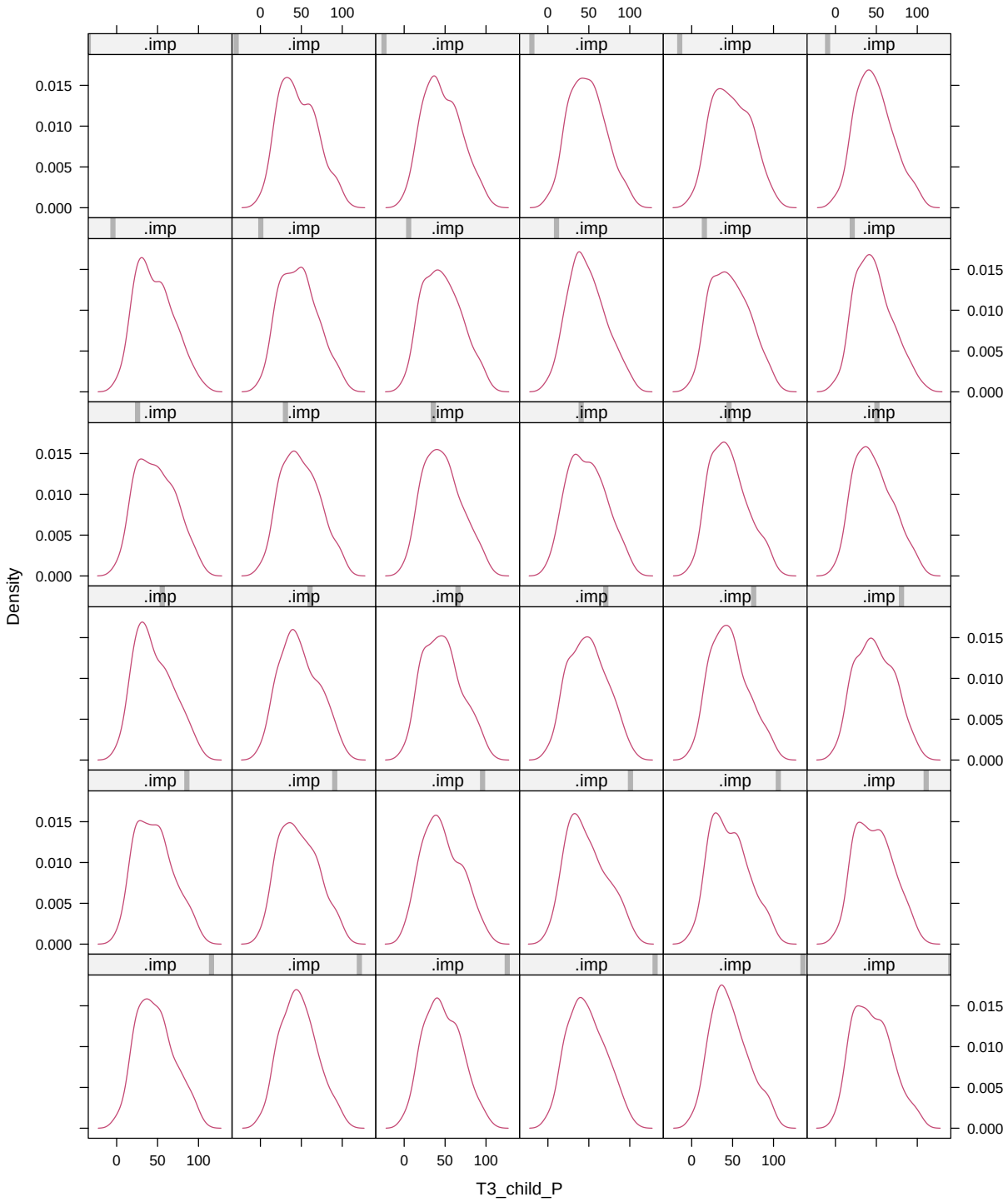
PSB T1



PSB T2

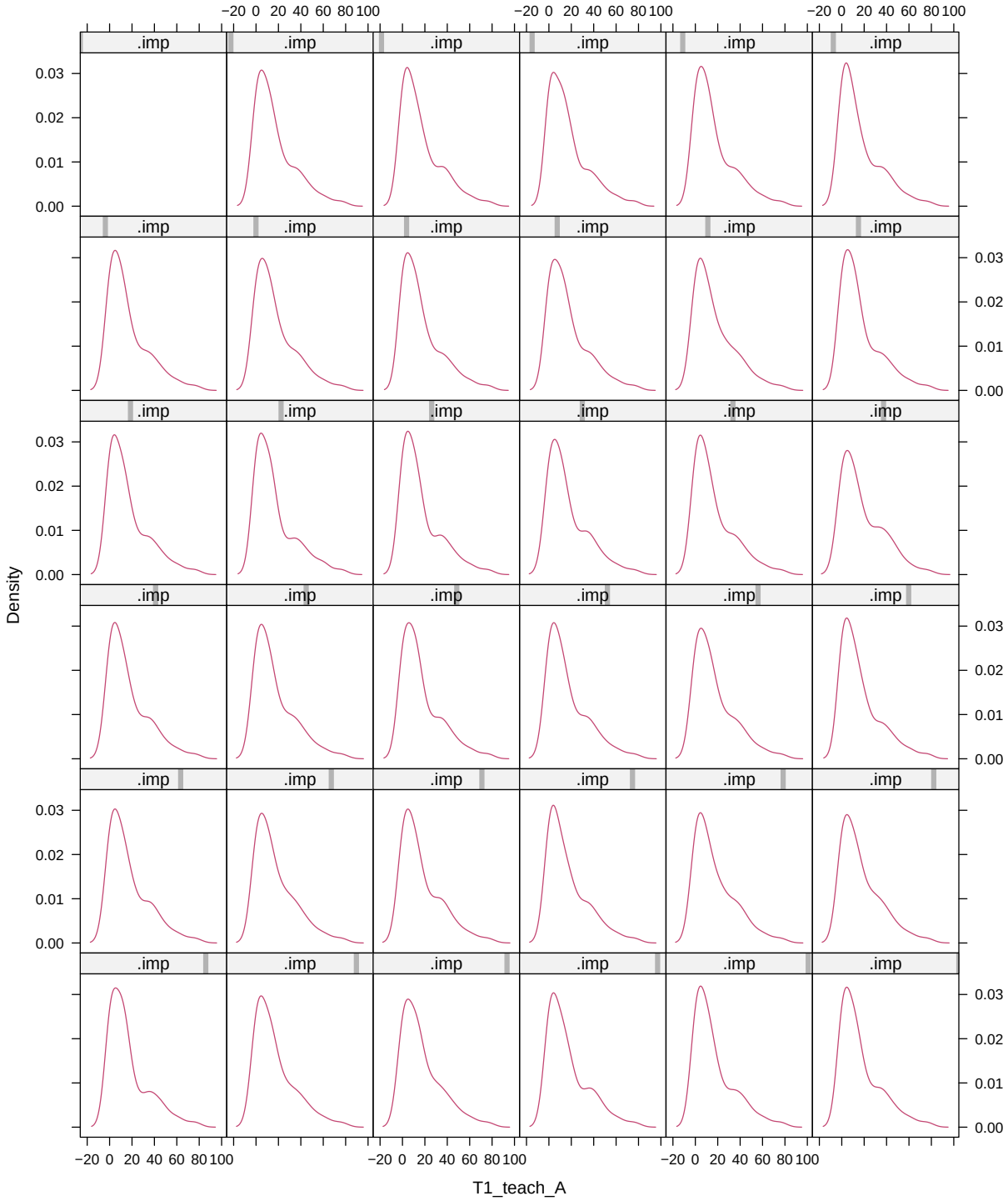


PSB T3

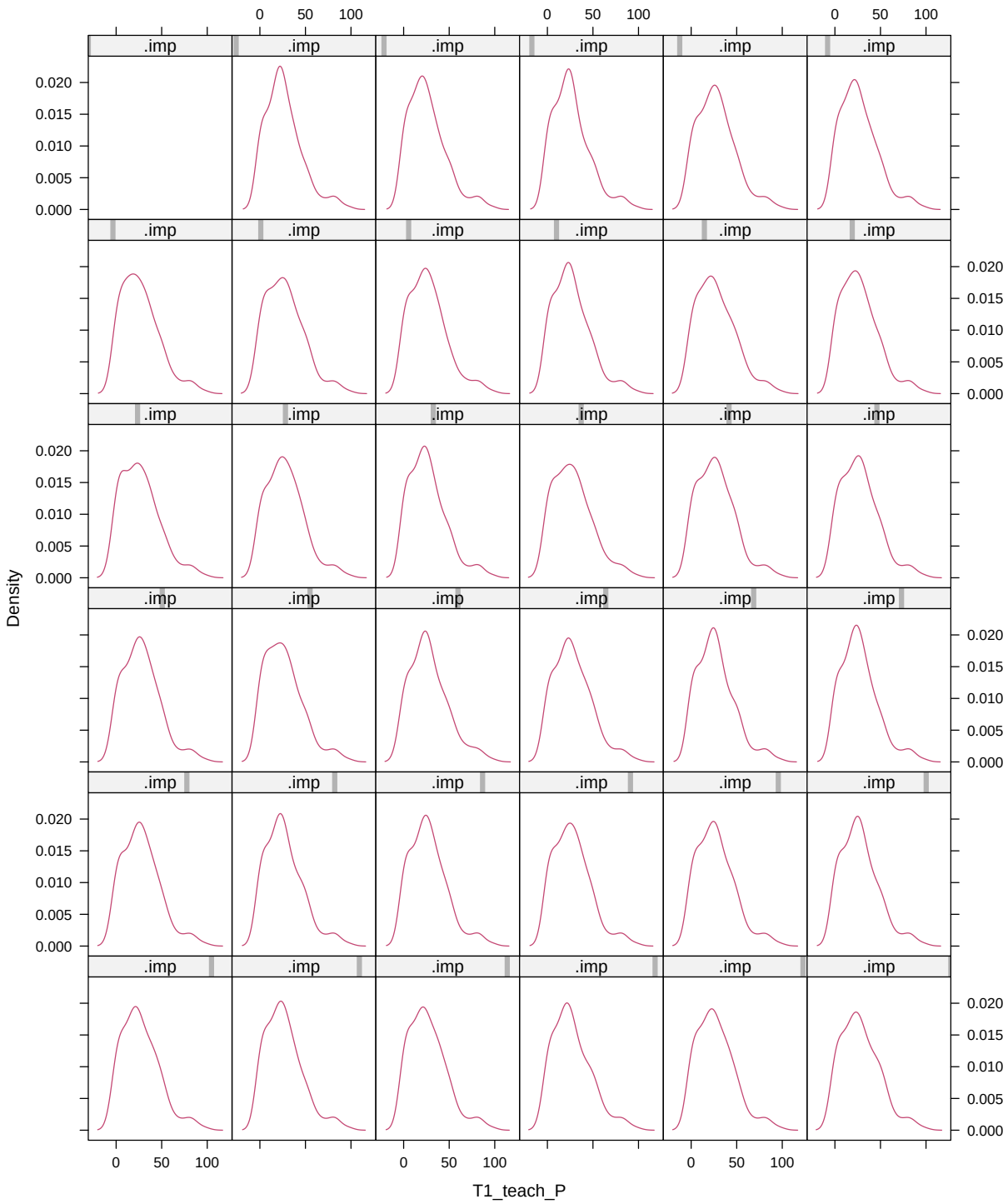


2) Teacher sumary scores for each timepoint

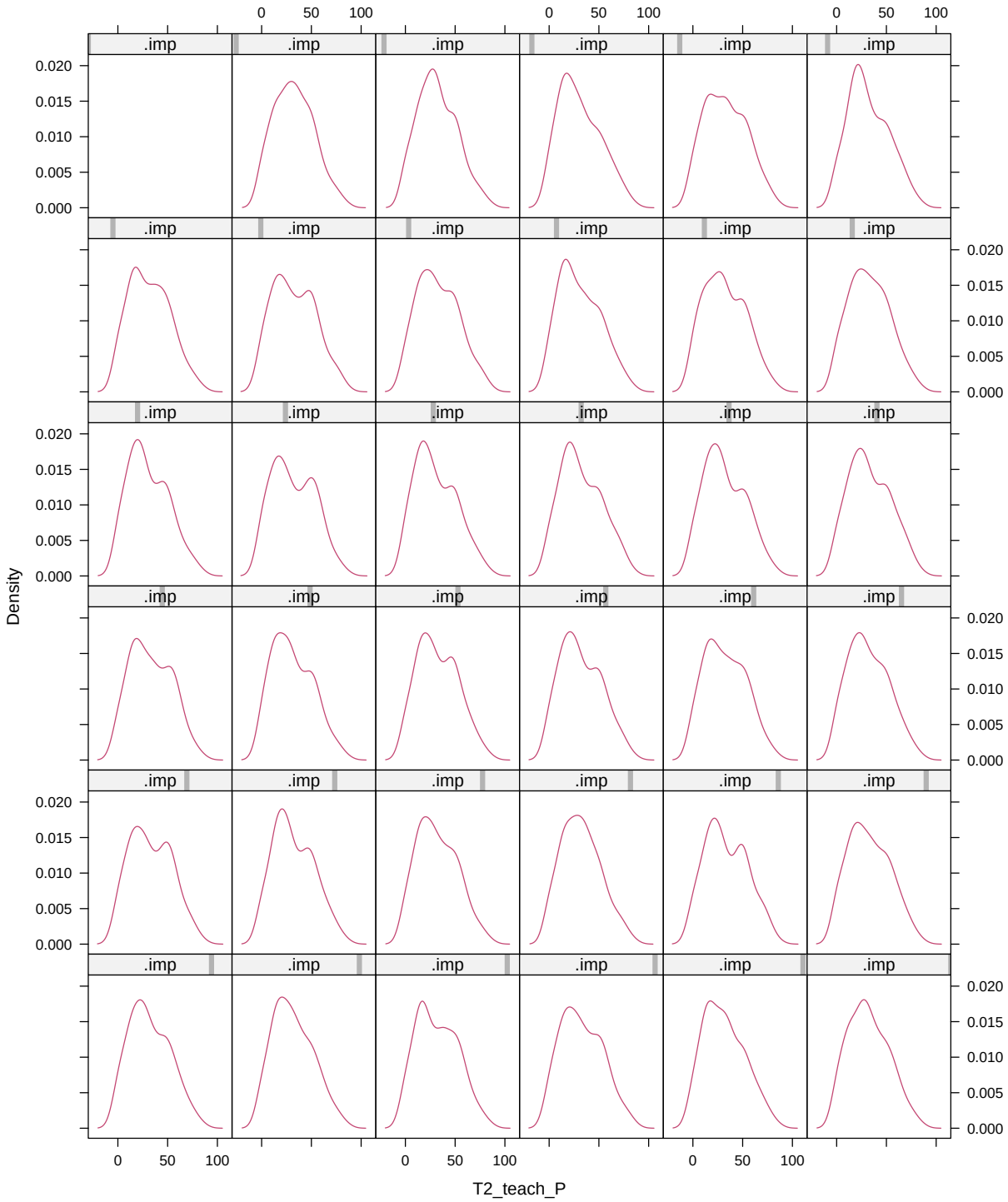
ASB T1



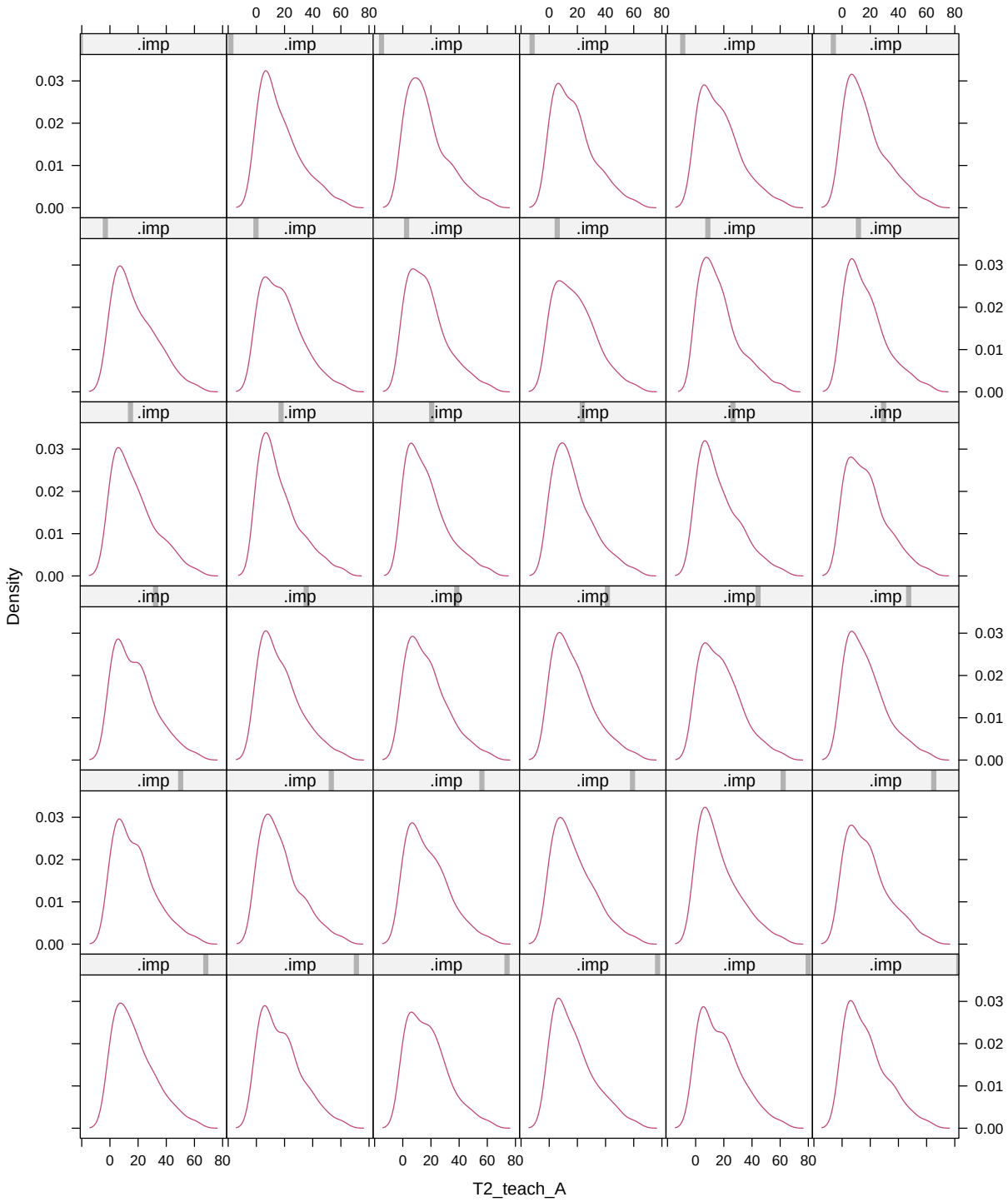
ASB T2



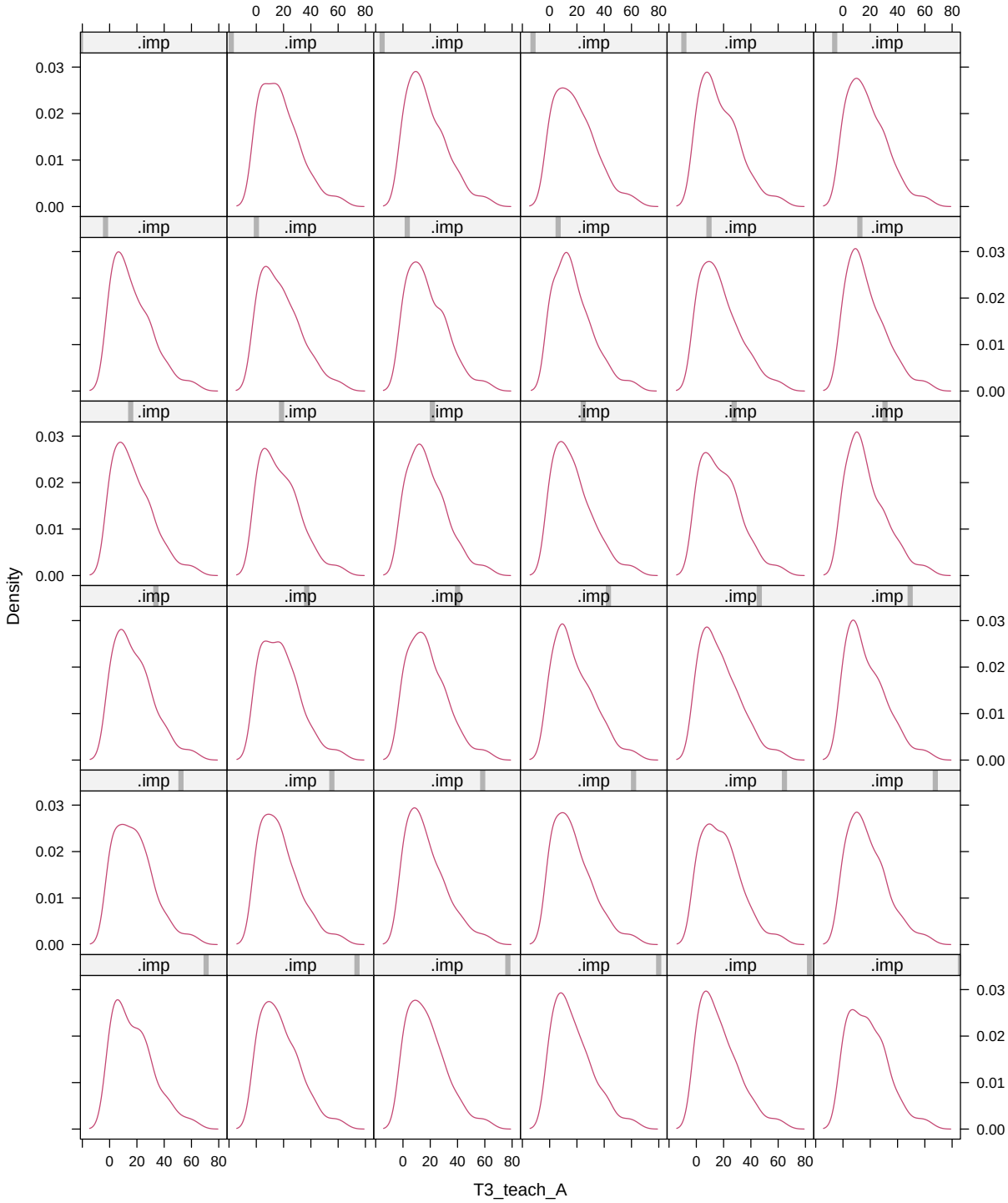
ASB T3



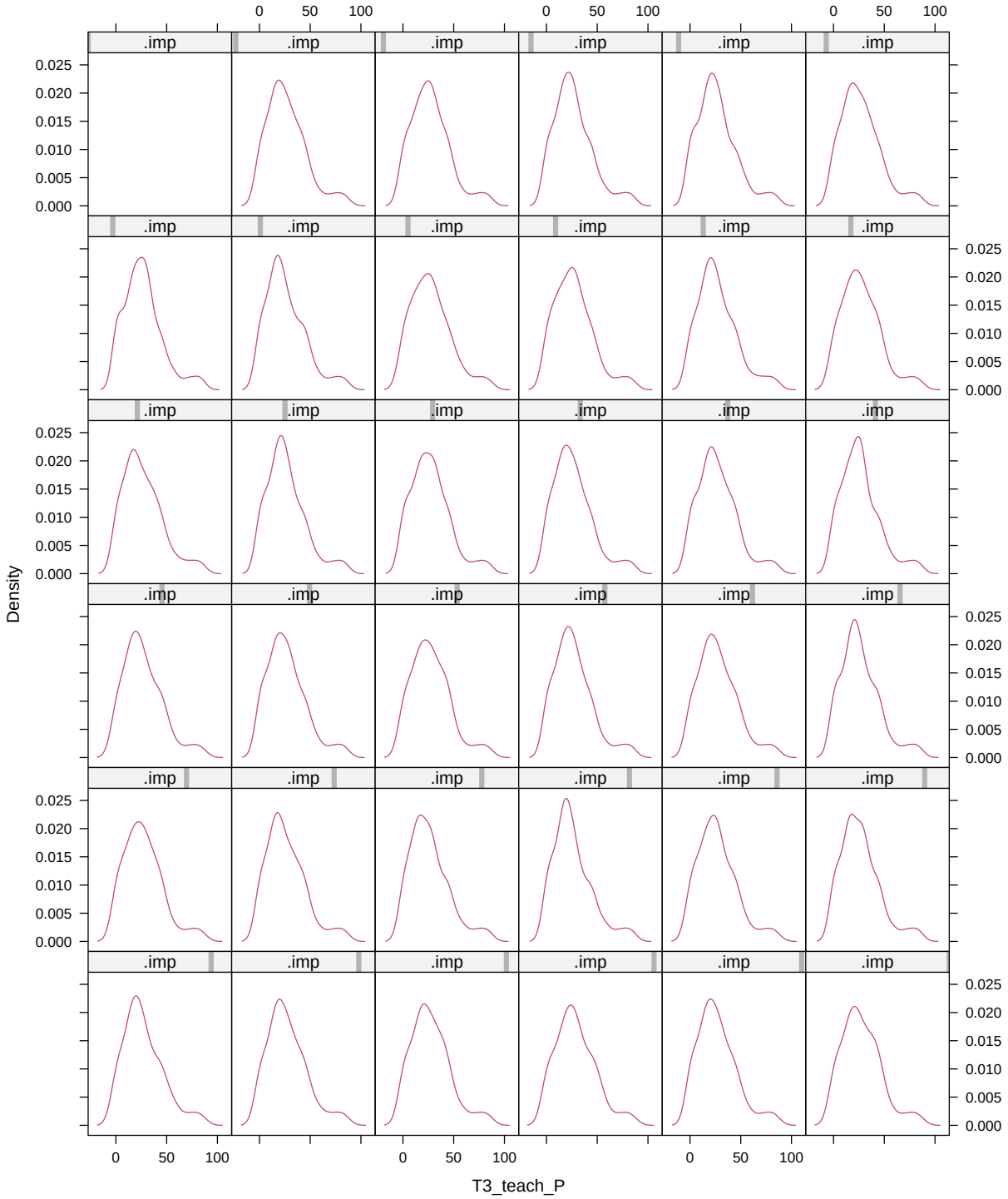
PSB T1



PSB T2

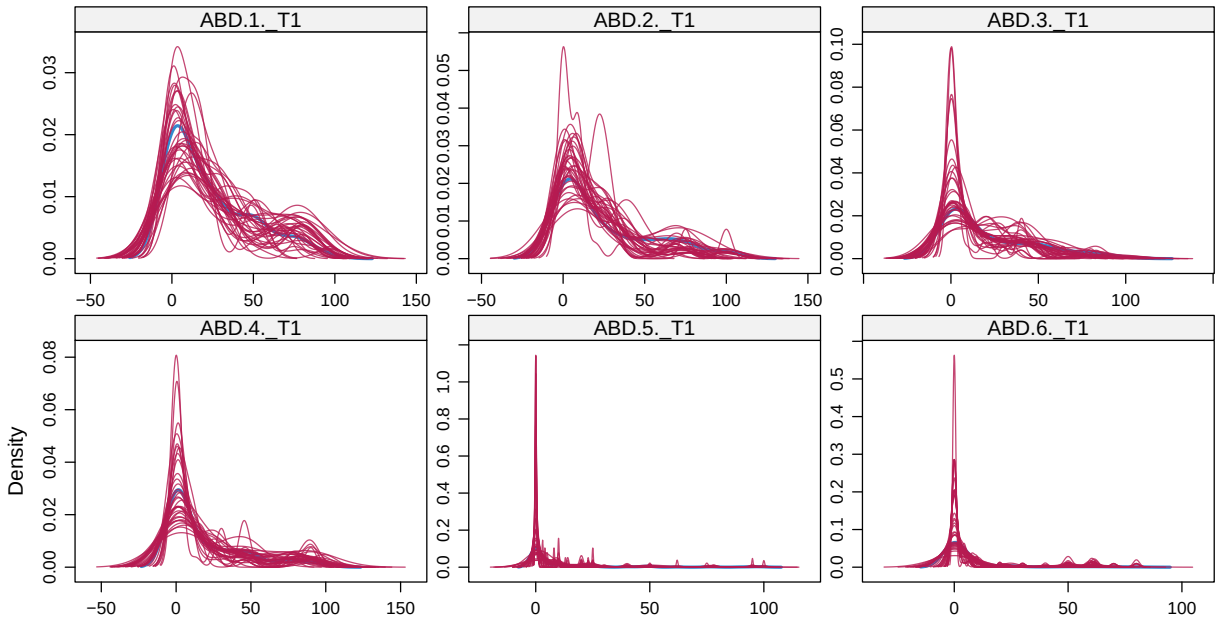


PSB T3

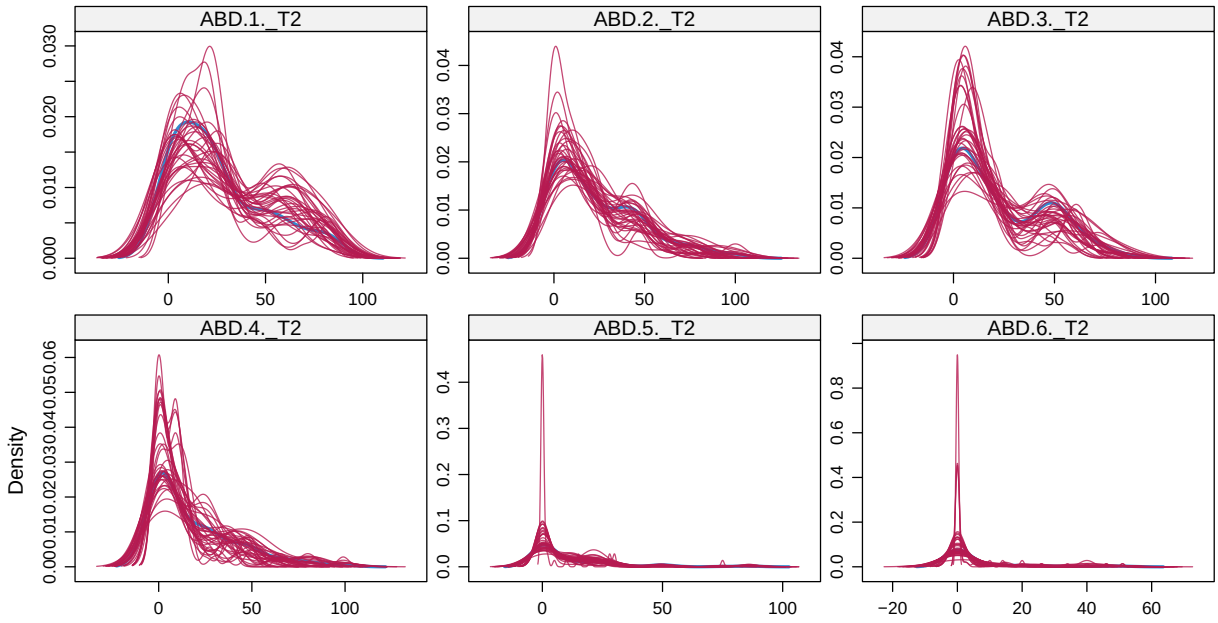


3) Kernel density plots comparing observed and imputed distributions for the individual items comprising each scale

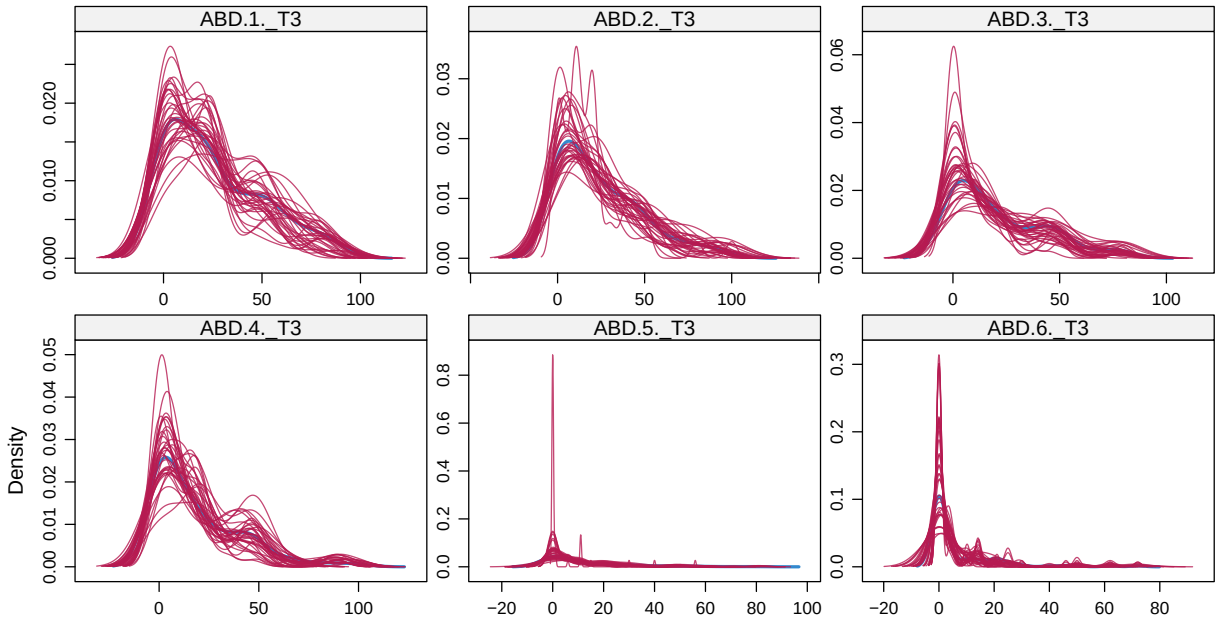
1) Teach ASB D1



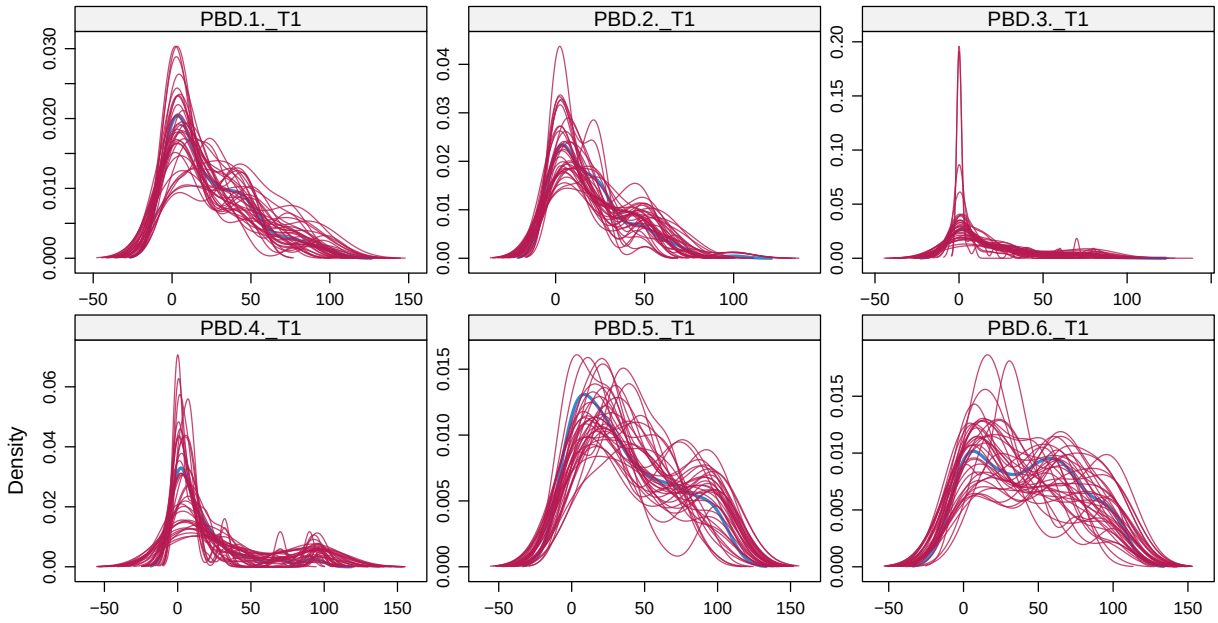
2) Teach ASB D2



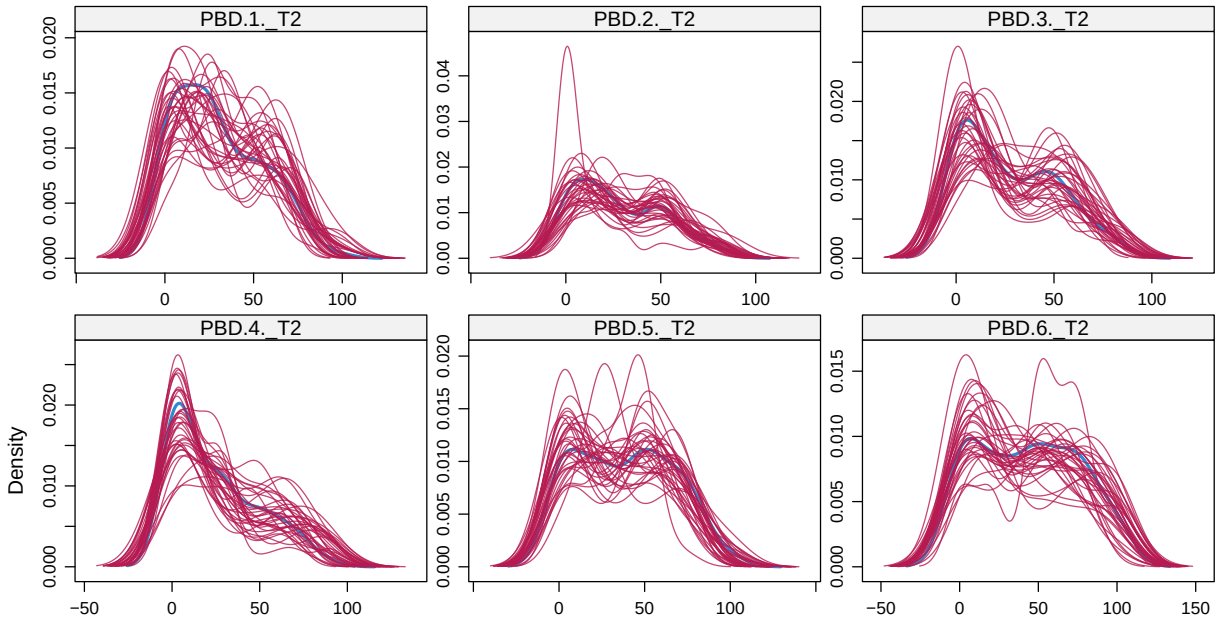
3) Teach ASB D3



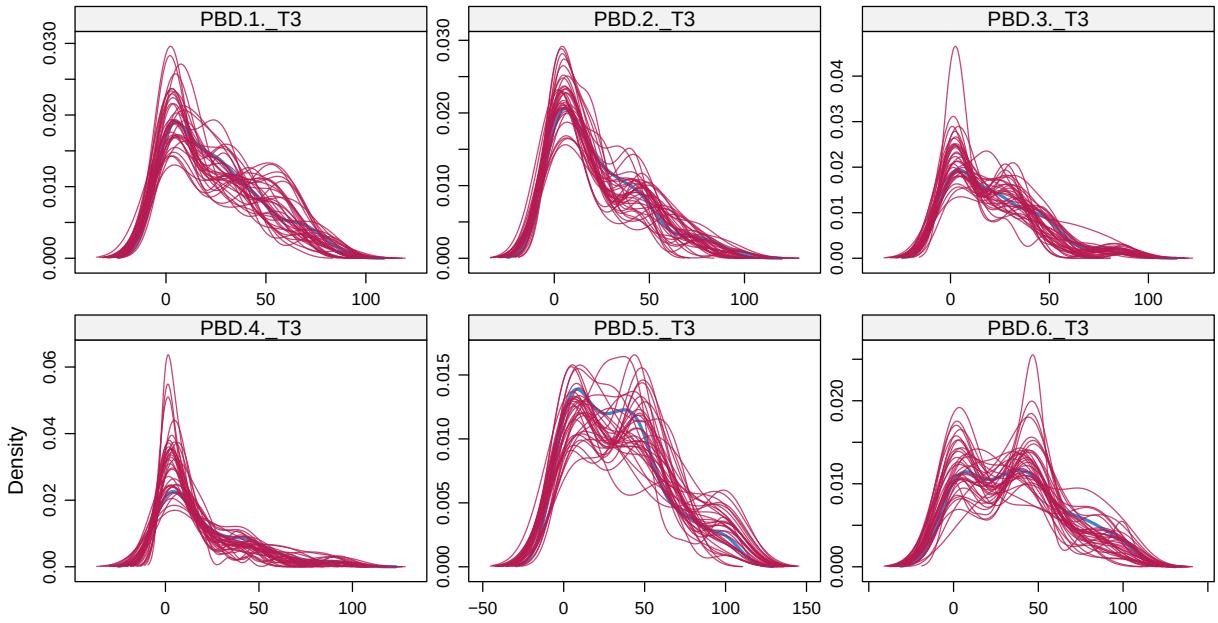
4) Teach PSB D1



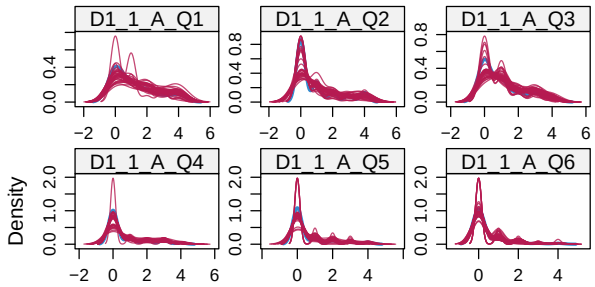
5) Teach PSB D2

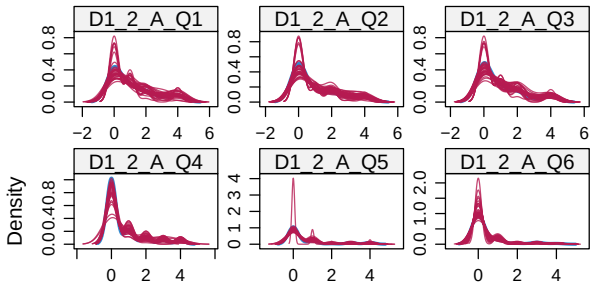


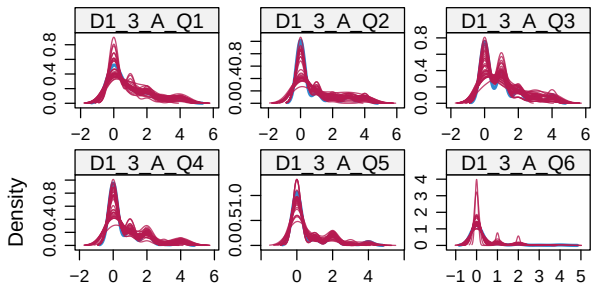
6) Teach PSB D3

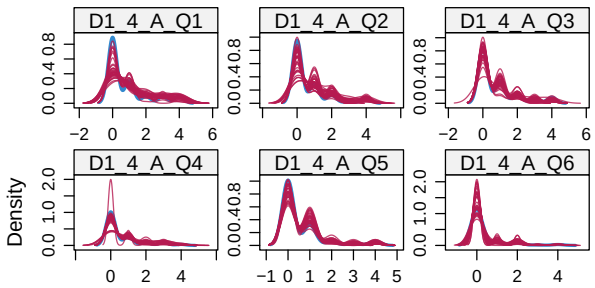


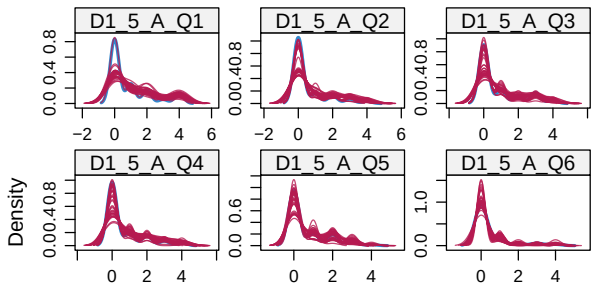
7) Child ASB D1



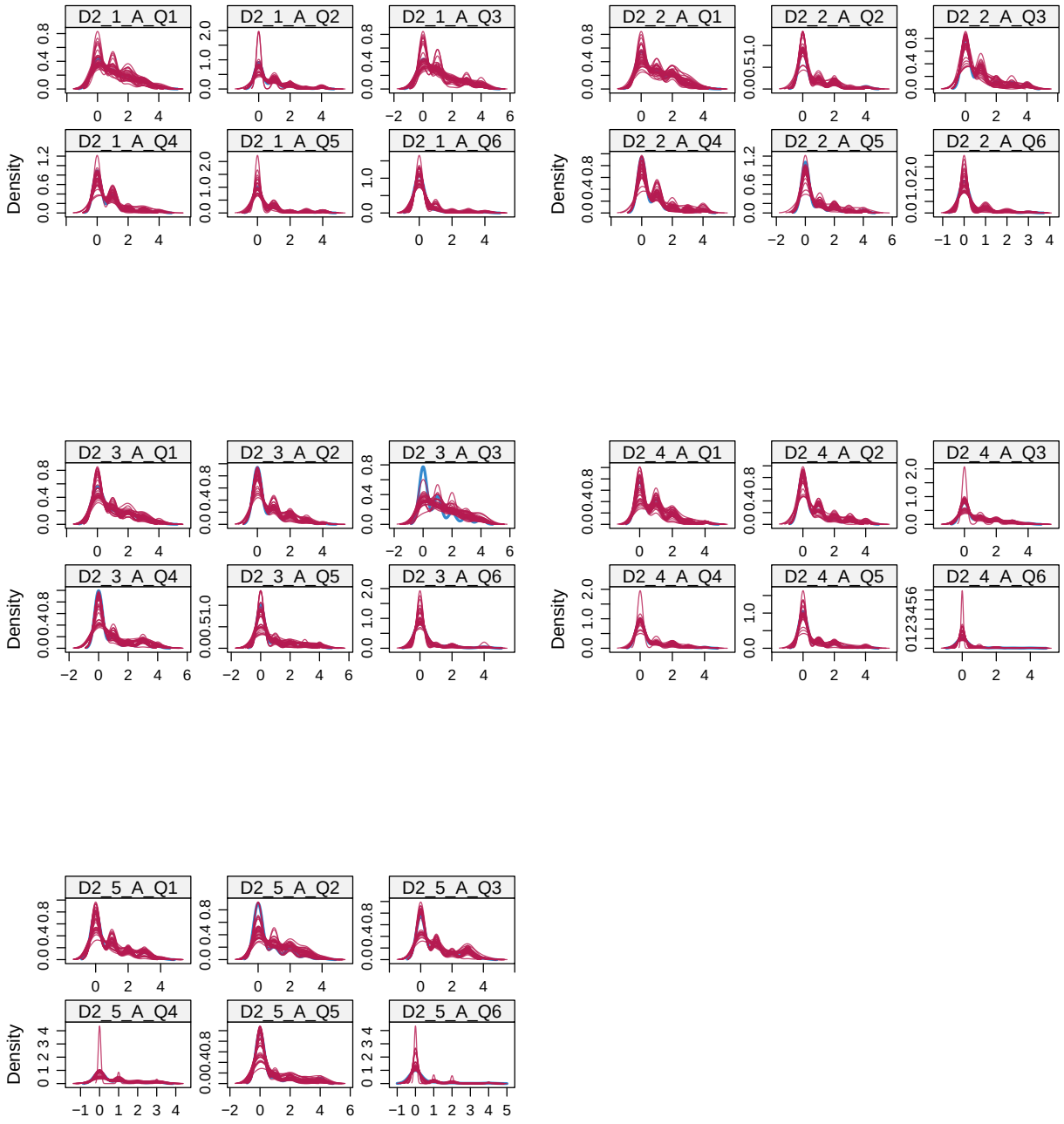




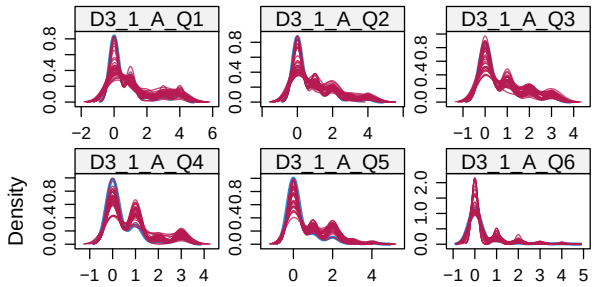


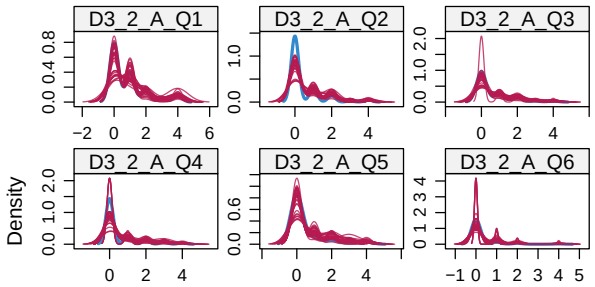


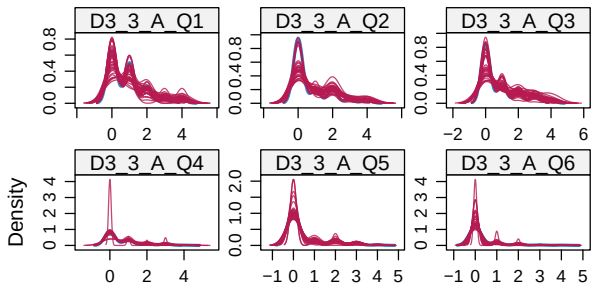
8) Child ASB D2

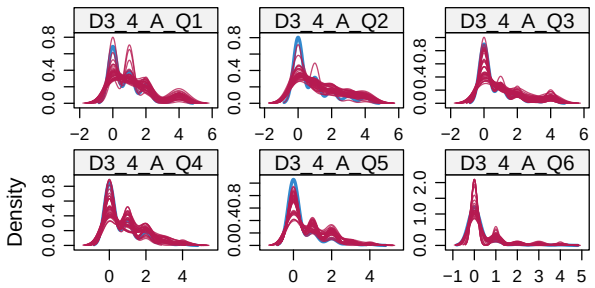


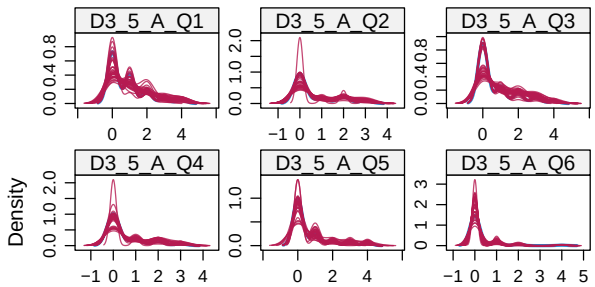
9) Child ASB D3



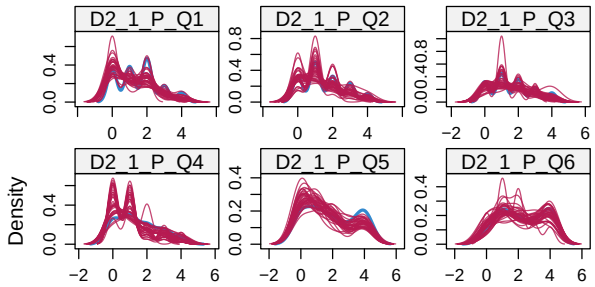


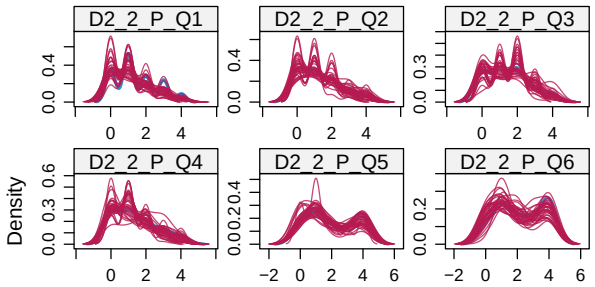


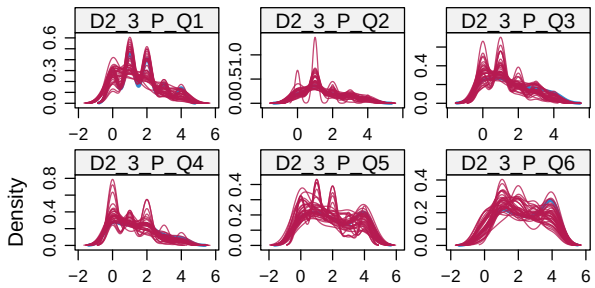


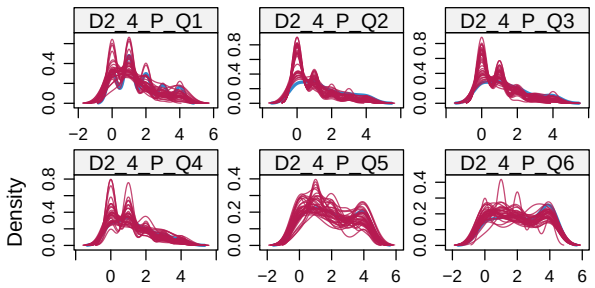


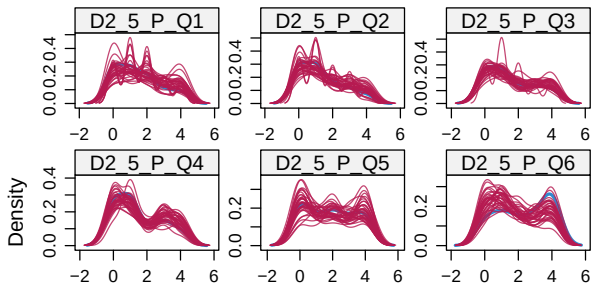
10) Child PSB D1



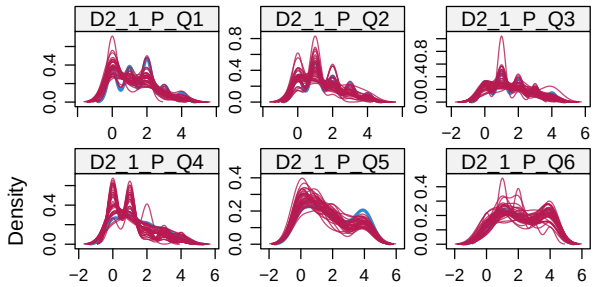


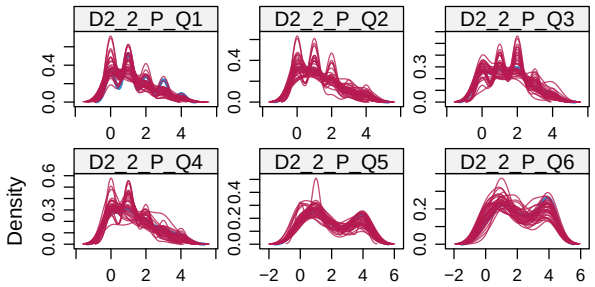


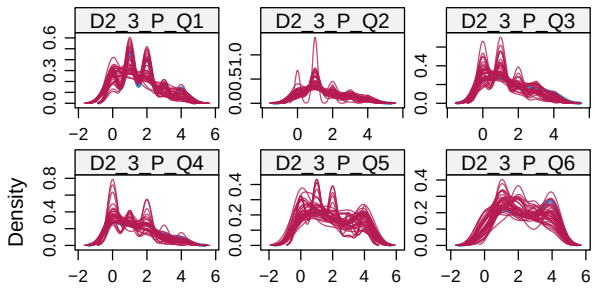


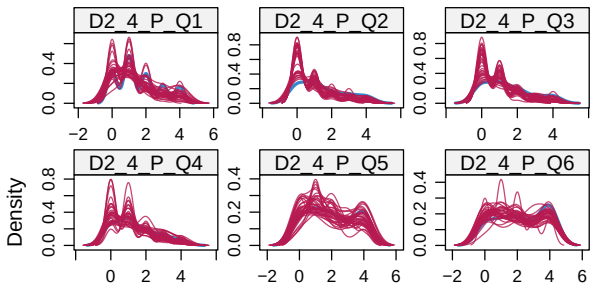


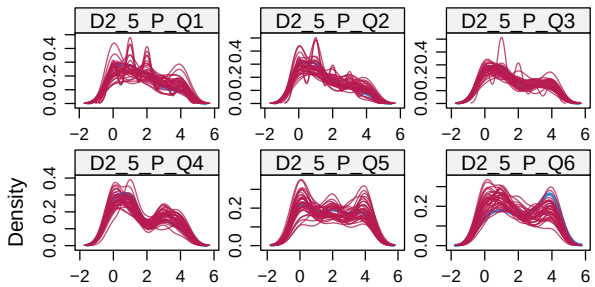
11) Child PSB D2











12) Child PSB D3

